ZULFARIDA

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PROFESSIONAL SUMMARY

Informatics Engineering student (Full Stack Developer track, CEP-CCIT FTUI × State Polytechnic of Jakarta) with hands-on experience in systems analysis, multi-platform software project coordination, **end-to-end quality assurance, and technical documentation** — demonstrated through real-world projects including CuitBox (1st Winner, CCIT Venture Studio Batch #1) and a multi-platform Coffee Shop Information System (Undergraduate Thesis).

Equipped with a full-stack technical background that enables deep understanding of system architecture, API flows, database structure, and software lifecycle — allowing me to design precise test scenarios, communicate technical issues clearly to developer teams, and identify system risks early. AI-assisted workflow practitioner with proven ability to accelerate QA documentation, test planning, and bug triage processes.

Proven cross-functional collaborator with 2 years of organizational leadership experience. Open to opportunities in **Software Quality Assurance (QA)** supporting digital transformation initiatives in enterprise-scale environments such as Telco, BFSI, and Utility sectors.

ACHIEVEMENTS & AWARDS

- 1st Winner, CCIT Venture Studio Batch #1 "Manggala" – Startup Incubation Program, CCIT FTUI (2026) (Awarded as Co-Founder, Hacker (Engineer), and Hipster (UX Designer) of CuitBox an integrated photobooth management system adopted by 40+ real users and validated by 23 User Acceptance Testing (UAT) respondents across web booking, admin management, and desktop kiosk platforms.

TECHNICAL SKILLS

- Quality Assurance & Testing: Black-Box Testing, Functional & Integration Testing, End-to-End (E2E) Cross-Platform Testing, Test Case & Scenario Design, API Testing, Regression Testing & Bug Triage, Hardware-Software Integration Testing, Edge-Case & Reliability Testing, Payment Transaction Flow Testing
- Business & Systems Analysis: IT Business Architecture Analysis, System Architecture Mapping, Information Systems Integration, Technical Documentation
- Software Engineering Practices: Software Quality Assurance (SQA) & Testing, Human-Computer Interaction (HCI) Design, Object-Oriented Programming (OOP), DevOps Workflow & Project Lifecycle Awareness
- Databases & Information Systems: MySQL (Schema Design & Data Flow Reading), Management Information Systems (MIS)
- Security & Infrastructure: Digital Signature & PKI, X.509 Certificate Authentication, SSL/TLS Protocol Analysis, Linux (CentOS) & Windows Server Administration
- Programming Languages (Supporting Knowledge): Java, Python, PHP, Dart, JavaScript
- Frameworks & Platforms (Supporting Knowledge): Spring Boot, Flutter, CodeIgniter 4
- Tools & Version Control: Git, GitHub (21 public repositories, 200+ contributions in the past year)
- Networking: Cisco CCNA – Introduction to Networks (Cisco Networking Academy)
- Digital Marketing: AI-Assisted Copywriting for Digital Advertising, Social Media Management for Digital Brands

PROFESSIONAL SKILLS

- Analytical Thinking & Attention to Detail — Identifying system gaps, edge cases, and inconsistencies across multi-platform environments
- Cross-Functional Team Coordination & Stakeholder Communication — Bridging developer teams, product managers, and business stakeholders in bug triage and release validation
- Initiative, Ownership, and Proactive Problem-Solving — Independently designing test scenarios and identifying risks before they escalate
- Structured Documentation & Reporting — Clear test case writing, bug reporting, and QA progress communication to technical and non-technical audiences

- Organizational Leadership & Cross-Divisional Collaboration — Proven ability to coordinate across sub-teams, manage timelines, and ensure quality across parallel workstreams
- Mentoring, Training, and Peer Guidance

KEY PROJECTS

CCIT Venture Studio Batch #1 "Manggala" - Startup Incubation Program | *6-Month Program, CCIT | 2026*

Selected participant in a 6-month startup incubation program for CCIT students and 1st Winner of Batch #1 (Final Score: 74.0), building a real digital product from concept to launch as part of a founder-style startup team.

CuitBox (Photobox Platform) - Self-Service Photo Booth Platform (Web & Desktop Kiosk) *|Co-Founder | Hacker (Engineer) & Hipster (UX Enginer) | Full-Stack Developer & QA / Core Backend Architect*

- Co-founded CuitBox and contributed to full-cycle product development from concept through launch, resulting in product adoption by 40+ real users and structured evaluation by 23 User Acceptance Testing (UAT) respondents
- Conducted end-to-end (E2E) functional, integration, and user-journey testing across the customer booking web, admin dashboard, and desktop kiosk applications
- Performed reliability and edge-case testing for hardware-software integration (DSLR/webcam feed stability, automated background recording) and transaction handling (Midtrans QRIS callback & payment timeouts)
- Worked across the booking and kiosk sub-teams, resolving merge conflicts and cross-platform bugs to maintain release stability
- Hacker (Engineer): Developed real-time camera live view integration for DSLR/webcam and built the desktop kiosk application with automated background video recording
- Hacker (Engineer): Developed the website admin dashboard covering reservation management, real-time slot scheduling, dynamic pricing, booth machine management, and financial report exports (Excel/CSV/PDF)
- Designed the relational database architecture and built RESTful APIs connecting the web booking platform and the desktop kiosk application
- Implemented Role-Based Access Control (RBAC) and a secure authentication system, including login, remember-me, and password recovery
- Integrated third-party services: Midtrans QRIS payment gateway, Google Drive OAuth for photo archiving, and automated SMTP email delivery
- Hipster (UX/Designer): Designed intuitive UI/UX wireframes in Figma for the website admin panel and created customizable photo frame templates

Repositories: Photobox_Website, photobox-dekstop

Capstone Project & Undergraduate Thesis Web and Mobile-Based Coffee Shop Management System Using CodeIgniter 4 and Flutter Frameworks *4th Semester | Score: 80 (A-) | 2025–2026*

Multi-platform integrated coffee shop management system comprising a customer web application (CodeIgniter 4), staff mobile POS (Flutter), and REST API backend (Spring Boot).

- Conducted system requirements analysis and mapped user needs for customer, staff, and admin personas
- Designed multi-role access control architecture and overall system data flow
- Designed MySQL database architecture and Spring Boot REST API contracts across all platforms
- Validated system functionality through automated cross-platform testing (Black-Box Testing)
- Implemented dynamic QR payment integration and custom order workflows
- Documented system architecture, test scenarios, and validation results in formal undergraduate thesis report

Repositories: ta-ci4-web-coffeeshop, ta-flutter-aplikasi-coffeeshop, ta-server-coffeeshop

Group Academic Projects

Testing for Coffee Shop Web Application *(Software Quality Assurance Project)* | *4th Semester | Score: 76 (B+)*

- Designed structured black-box test cases and test scenarios for a full coffee shop web application
- Performed functional, integration, and regression testing across all system modules
- Identified, documented, and classified defects by severity in formal bug reports
- Produced complete QA documentation: test plan, test case matrix, and test result report

Coffee Shop Web Application (Web Application Project) | *3rd Semester | Score: 88 (A)*

- Developed and validated a full-featured coffee shop web application covering UI, backend logic, and database integration
- Conducted functional validation across all system modules and user flows

Coffee Shop – Enterprise Information System (Developing Enterprises Information System using Framework Project) | *3rd - 4th Semester | Score: 78 (B+)*

- Built and validated an enterprise-scale web information system for coffee shop operations using structured MVC framework
- Applied software quality practices in structured development and functional validation

Coffee Shop Mobile Application (Mobile Programming Project) | *4th Semester | Score: 81 (A-)*

- Developed and tested a mobile application with cross-device functionality validation
- Conducted mobile UI and functional testing across device configurations

Digital Signature & X.509 Certificate Authentication Analysis in SSL/TLS (Security Concept) | *3rd Semester | Score: 86 (A)*

- Analyzed the implementation of Digital Signature and X.509 Certificate Authentication for server authentication in the SSL/TLS protocol
- Applied security testing principles to evaluate protocol integrity and authentication flow
- Produced formal analysis report documenting findings and security implications

Todo List Application – HCI Redesign (Human Computer Interaction Project) | *3rd Semester | Score: 84 (A-)*

- Applied HCI principles and usability evaluation to redesign and validate a task management application
- Conducted user journey analysis and usability testing to identify friction points and validate improvements

Java Application for Movie Theatre (Object Oriented Programming Project) | *2nd Semester | Score: 80 (A-)*

- Developed a Java-based movie theatre booking application using Object-Oriented Programming principles
- Demonstrated understanding of class structure, inheritance, and system logic flow — foundational for reading and understanding developer code during QA

Coffee Shop – Algorithm and Data Structure (Algorithm and Data Structure Project) | *1st Semester | Score: 89 (A)*

- Implemented and validated data structure algorithms with structured test case verification
- Demonstrated strong logical thinking and structured problem-solving

DevOps Implementation for Coffee Shop Website (Project Management using DevOps Project) | *3rd Semester | Score: 64 (C+)*

- Applied DevOps workflow concepts to a coffee shop web application project

IT Implementation for a Rental PlayStation Business (Introduction to Information Technology Project) | *1st Semester | Score: 64 (C+)*

- Analyzed IT implementation requirements and business process needs for a small PlayStation rental business
- Produced IT solution recommendations based on operational requirements

Business Architecture Analysis of Apple Inc. (Information System Architecture and Technology Project) | *1st Semester | Score: Score: 71 (B-)*

- Analyzed strategic IT-business alignment of Apple Inc. using business architecture frameworks
- Produced formal IT business strategy analysis documentation

Integration of Management Information Systems in Manufacturing (Management Information Systems) | *2nd Semester | Score: 57 (C)*

- Studied and documented MIS integration patterns in a manufacturing environment

Manufacturing Information System at Indomie Company (Management Information Systems) | *2nd Semester | Score: 72 (B-)*

- Analyzed and documented the Manufacturing Information System implementation at Indomie Company

Linux CentOS Server Configuration (Operating System) | *1st Semester | Score: Score: 72 (B-)*

- Configured and administered a Linux CentOS server environment

Windows 11 Main Memory Management (Operating System) | *1st Semester | Score: Score: 72 (B-)*

- Implemented distributed system concepts using Spring framework for a coffee shop web application

Website for Travel Destination (Introduction to Web Programming Project) | *2nd Semester | Score: 74 (B-)*

- Developed a travel destination website covering frontend and basic backend functionality

Cinema Database Design (NoSQL - MongoDB Fundamentals) | *2nd Semester | Score: 69 (D+)*

- Designed and implemented a cinema database management system

Independent Projects (GitHub)

BISINDO Gesture Recognition System v2.0

Python-based real-time Indonesian Sign Language (BISINDO) alphabet translator using Computer Vision (MediaPipe) and Machine Learning (Random Forest + SVM Ensemble).

- Designed and implemented a cinema database management system
- Developed and validated a 156-dimension feature extraction pipeline (hand landmark coordinates + finger joint angles) for classifying 26 BISINDO letters
- Trained and tested an Ensemble ML model (Random Forest 300 estimators + SVM RBF with Soft Voting) with in-memory data augmentation for robustness across camera angles and distances
- Implemented and validated Text-to-Speech (TTS) output via Windows SAPI & Google TTS API, ensuring zero-crash reliability on repeated voice calls
- Designed offline-first architecture: 100% local processing with no video data transmitted externally (privacy-by-design)

Stack: Python, MediaPipe, Scikit-Learn, OpenCV, Windows SAPI, gTTS, Pygame

Stellar Task Tracker Dapp

Decentralized task management system built on the Stellar blockchain using Soroban SDK and Rust smart contracts.

- Designed offline-first architecture: 100% local processing with no video data transmitted externally (privacy-by-design)
- Decentralized task management system built on the Stellar blockchain using Soroban SDK and Rust smart contracts.
- Designed and deployed a smart contract on Stellar's Soroban network for secure, immutable task management (create, retrieve, complete, delete)
- Validated smart contract function behavior: add_task(), get_tasks(), complete_task(), remove_task() — ensuring data integrity and correct state transitions on-chain
- Implemented persistent blockchain storage with unique ID generation and category-based task classification

Stack: Rust, Soroban SDK, Stellar Blockchain

ORGANIZATIONAL & LEADERSHIP EXPERIENCE

Human Resource Development (PSDM) Member | *University of Indonesia Robotics Team (TRUI) | 2024 - 2025*

- Managed internal and external communications among team members
- Organized internal initiatives including benchmarking visits, member inaugurations, and development programs

Master of Ceremony | *University of Indonesia Robotics Team (TRUI) x Mechanical Autonomous Navigation Control (MAUNC) from Siliwangi University (UNSIL) Robotics Benchmarking Event | 2024*

- Hosted a cross-institution technical exchange event, ensuring structured session transitions and audience engagement

Logistics & Medical Division Staff | *Computer Science Festival (CSFEST) 2024 Committee | 2024*

- Supported event logistics and provided on-site medical readiness during a large-scale campus technology festival

Greeter & Volunteer | *D'Futuro Futurist Summit 2024 | 2024*

- Managed event access control and ticketing operations; welcomed attendees and provided facility information

Public Relations Staff | *Android Community CCIT FTUI (KOMANDRO) | 2024 - 2025*

- Managed internal and external communications and built relationships with external partners to support organizational activities

Master of Ceremony | *Android Community CCIT FTUI (KOMANDRO) Chairman Election | 2024 - 2025*

- Facilitated structured proceedings from opening to closing, coordinating with committee members and speakers

Mentor | *Masa Bimbingan (MABIM) Freshman Orientation Program | 2025*

- Coordinated cross-generational event planning, participant logistics, and technical requirements

Public Relations Committee | *Android Community CCIT FTUI (KOMANDRO) Alumni Reunion, 2013 - 2025 Cohorts | 2026*

- Coordinated cross-generational event planning, participant logistics, and technical requirements

Event Committee Member, SELARAS (Semarak Kepalangmerahan Indonesia) | *Indonesian Red Cross Youth (PMR), SMA Negeri 12 Kota Depok | 2024*

- Coordinated cross-generational event planning, participant logistics, and technical requirements

Member | *Indonesian Red Cross Youth (PMR), SMA Negeri 12 Kota Depok |* 2023 *- 2024*

EDUCATION

Diploma in Informatics Engineering - Full Stack Developer Track | *2024 - Present (5th Semester)*

State Polytechnic of Jakarta *in collaboration with Continuing Education Program - Center for Computing and Information Technology (CEP - CCIT FTUI), Faculty of Engineering, Universitas Indonesia*

Senior High School (SMA) | *SMA Negeri 12 Kota Depok - Science Track (IPA) | Graduated 2024*

CERTIFICATIONS

- CCNA: Introduction to Networks - Cisco Networking Academy, State Polytechnic of Jakarta (2025)
- Smart Digital Finance: Managing Business Finance in the Technology Era - m-ARTHAMEDIA & KasirPintar (2026)
- From Campus to Industry Reality - CEP-CCIT FTUI (2025)
- Copywriting AI for Digital Advertising - Digital Talent Scholarship, Ministry of Communication and Digital Affairs (Komdigi) (2025)
- Social Media Management for Digital Brands - Digital Talent Scholarship, Ministry of Communication and Digital Affairs (Komdigi) (2025)
- Mandarin Language Course, Beginner Level - Brilliant English Course (Brilliant EC) (2024)
- English Course, Beginner Level - The Eagle English Course (2022)